MANOJ KUMAR G

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PROFESSIONAL SUMMARY

Computer Science graduate (PES University, 2026) looking for data, business and BI analyst roles. Built an end-to-end analytics project on 99,441 e-commerce orders, from a PostgreSQL warehouse and 18 SQL queries to a statistical test, a demand forecast and a four-page Power BI dashboard. As a research intern, developed an explainable ML pipeline on Indian stock data. Compared every model with a simple baseline before reporting results.

SKILLS

SQL & Databases: SQL (joins, CTEs, window functions), PostgreSQL, MySQL, star-schema data modelling
BI & Visualization: Power BI (DAX, Power Query), Advanced Excel (Pivot Tables, VLOOKUP/XLOOKUP), Tableau
Data Processing: Python (pandas, NumPy), ETL pipelines, data cleaning, validation, data-quality tests, reconciliation
Analytics & Statistics: Hypothesis testing (Mann-Whitney U), cohort, RFM and Pareto analysis, root-cause analysis, time-series forecasting (Holt-Winters), walk-forward backtesting, KPI reporting
Machine Learning: scikit-learn, XGBoost, TensorFlow/Keras, PyTorch, CNN/LSTM, SHAP, FinBERT, OpenCV, Optuna
Tools & Development: Git, GitHub Actions (CI/CD), Docker, Jupyter, Flask, Streamlit, Jira, Agile, SDLC

PROJECTS

E-Commerce Revenue & Customer Retention Analytics | Personal Project **2026**
Python, SQL, PostgreSQL, Power BI, DAX, Docker, GitHub Actions | github.com/MARCUS-00/ecommerce-retention-analytics

- Cleaned and modelled the public Olist dataset (99,441 orders, nine tables, 2016–2018) into a PostgreSQL star schema with a Python ETL pipeline, 10 automated data-quality tests in GitHub Actions CI and a data dictionary.
- Wrote 18 SQL queries using joins, CTEs and window functions: 15 answer business questions (revenue trend, category Pareto, cohort retention, RFM segments) and 3 run data-quality and reconciliation checks.
- Found that only 3.0% of 93,358 customers placed a second delivered order and 8.11% of delivered orders arrived late; a Mann-Whitney U test showed late orders had a median review of 2/5 vs 5/5 on time ($p < 0.001$).
- Built a four-page Power BI dashboard with 43 DAX measures and summarised the findings and recommendations in a written executive memo for non-technical readers.
- Backtested a Holt-Winters demand forecast against a seasonal-naive baseline over six walk-forward folds; it did not beat the baseline (WAPE 0.2310 vs 0.2308), so recommended the simpler model.

Early Detection of Osteoporosis from Dental X-rays (OsteoScan) | Team Capstone, PES University **2025 – 2026**
TensorFlow/Keras, EfficientNetB0, OpenCV, Flask, Docker, GitHub Actions | Live app: manojkumar724-osteoscan.hf.space

- Deployed a research-prototype Flask app on Hugging Face Spaces that sorts dental X-rays into three bone-density classes, with a Grad-CAM heatmap, a PDF report and a 14-test suite in GitHub Actions CI/CD.
- Traced an early 97.28% accuracy to patient-level leakage and switched to leave-one-source-out cross-validation (13 patients): 61.5% accuracy (95% CI 38.5–84.6%) and macro-F1 0.59, vs 53.8% accuracy for brightness alone.

WORK EXPERIENCE

Research Intern | C3I, PES University, Bengaluru **Jan 2026 – May 2026**

- Developed an explainable ML pipeline under faculty supervision to predict 10-day price direction for 40 Indian large-cap stocks, combining price and technical indicators, fundamentals, FinBERT news sentiment and macroeconomic events into 68 features.
- Engineered a per-stock sentiment-decay feature tuned on training data only, and guarded against lookahead bias with time-based splits, an automated leakage audit and a 17-test regression suite.
- Trained an AUC-weighted XGBoost and LSTM ensemble and reported its modest test AUC of 0.54 (0.50 is chance), noting that its accuracy sat below an always-UP baseline, rather than tuning for a better-looking number.
- Designed a regime-adaptive confidence threshold that produces a ranked daily watchlist, and backtested the ensemble signals with 0.1% transaction costs; returns trailed buy-and-hold (10.52% vs 12.23%), so no trading edge is claimed.
- Turned SHAP attributions into plain-English, per-prediction explanations in a Streamlit app for non-technical users.

EDUCATION

B.Tech, Computer Science | PES University, Bengaluru **2023 – Jul 2026**
Diploma, Computer Science | Dayananda Sagar Institute of Technology **2020 – 2023**